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Grounding Analysis Based on Different Soil Characteristics in The Distribution System

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***Abstract* – This study aims to obtain the type of grounding and resistance value in the distribution system of medium voltage UP3 South Surabaya ducts. In addition, to see the magnitude of the comparative value of the results of the resistance analysis stimulates ETAP. In the South Surabaya UP3 medium voltage distribution system. PT. PLN (Persero) improves and optimizes service to all of its customers. The grounding system at PT PLN UP3 South Surabaya. The existing PLN UP3 South Surabaya must comply with PUIL regulations of 2011, and the value of land holdings must be no greater than 5 Ω. Of the 15 grounding points analyzed, there are 3 points whose grounding values do not meet the requirements of PUIL 2011, namely the grounding values are still above 5 Ω. The smallest grounding resistance value before adding the rod electrode is 1.1 Ω at the 3rd point or BD 313 and after adding one rod and then paralleling it, the resistance value drops to 0.55 Ω. For the largest grounding resistance at the point that meets the requirements, namely 4 Ω at the 9th point or BD 653 and after adding one rod and then parallelizing it, the resistance value drops to 1.75 Ω. ETAPsimulation, the simulation results are affected by the value of x direction and the depth of the conductor. In addition, the simulation results are affected by the size of the rod. ETAP simulation results, the touch tolerable is 587.6 volts and the step tolerable is 1948.6 volts. For the largest grounding resistance at the point that meets the requirements, namely 4 Ω at the 9th point or BD 653 and after adding one rod and then parallelizing it, the resistance value drops to 1.75 Ω. ETAP simulation, the simulation results are affected by the value of x direction and the depth of the conductor.**

Keywords: Grounding Systems, Lightning Surge, Power Distribution

I. INTRODUCTION

Humans cannot be separated in terms of fulfilling the need for electrical energy every day. However, with the continuing growth of the population, especially in Indonesia, as recorded in BPS data in 2021, there are 2,880,284 people per square kilometer [1]. In this way, the amount of consumption and the need for electrical energy continues to increase, with an increase in 2021 of 5.82% compared to the previous year which amounted to 159.12 million. Seeing the facts that happened, PT. PLN (Persero) continues to strive to improve and optimize services for all of its customers as service providers in the electricity sector. Likewise, what is done by UP3 South Surabaya as part of the PLN stakeholders who continue to try to provide the best possible service for customers, especially in the South Surabaya area [2]. However, with recent weather changes that have become increasingly erratic due to the worsening effects of global warming, the operation of medium voltage networks is greatly influenced by natural factors, namely lightning which can cause surge voltages [2]. Lightning itself can occur due to the achievement of a sufficiently high charge value which results in electrical failure in the air that occurs in several parts of the atmosphere so that a large electrical charge shift arises [2]. The transition can occur in the cloud and from the cloud to the earth's surface.

The phenomenon that occurs at PT PLN (Persero) UP3 South Surabaya, which has a supporting pole for medium voltage overhead lines with concrete and wire construction, has a great potential for the pole to be struck by lightning either directly or indirectly. So, reliable security against it is important to note [3]. Lightning occurs when the charge in some parts of the atmosphere has an electric field that reaches a high enough value to cause electrical failure in the air resulting in a large transfer of electric charge. This charge transfer can occur in clouds, between clouds and from clouds to the earth's surface.

Lightning surge protection can be achieved in several ways, one of which is using a lightning rod (arrester) [4]. This lightning rod functions as a short circuit, when a lightning strike causes a surge, the lightning rod can channel the surge directly to the earth to avoid damage to electrical equipment and power outages in distribution lines [4]. The second option is to add a ground wire. The ground wire is routed along the feeder duct and the ground wire is routed to the highest point of the feeder line as a lightning protection line [4].

Therefore, to minimize damage to the feeder from direct and induced lightning strike disturbances, the distribution network below 20 kV must use a grounding conductor or no grounding conductor in the feeder area which is prone to lightning strikes [5]. Of the two safeguards above, there must be other supporting factors to function properly. This is none other than the existing grounding system must follow PUIL regulations of 2011 where the ground value must not be greater than 5 Ω [6]. Surges are difficult to ground directly and can cause

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dangerous protective devices such as lightning arrestors and grounding wires to fail, causing power outages in distribution lines [7].

Based on previous research [3] The problem that occurs is the distribution channel that uses radial-type voltage so it has the potential to be struck by lightning during high rainfall. At PT. PLN Persero UPJ. In Lamongan, there are several points in the grounding of the feeder distribution line that exceed 5Ω. This of course will have a big risk in an unsafe lightning surge so from the PLN side it requires an analysis to improve and minimize the value of the ground at the point of the grounding system [3].

Previous research [3] has disadvantages in terms of selecting the grounding method in multi-transformers because ground faults often occur, and each transformer has a large single-phase to ground fault current. If there is an error in the selection of transformer grounding, then this will cause serious damage to the device. Therefore, this research offers a grounding method calculation with the addition of ETAP simulation on the PT PLN (Persero) UP3 South Surabaya medium voltage distribution system.

A. Methodology Contents

In this study, only 15 out of 100 points of soil resistance were analysed. Because these 15 points often experience interference. Where the intended point is on each pole of the low voltage distribution channel on the Semolowaru feeder, and the distance from each pole is 55-60 meters.

Analysis was performed on the data obtained by manual calculation based on [15]. Figure 1. shows the flowchart and integrity of the scenarios for the analysis of this ongoing study. This can be seen from the beginning, in the name of analysis to obtain the value of grounding resistance by evaluating the structure of the electrodes to obtain a grounding value according to the type or type of soil in the Semolowaru feeder distribution channel. Then an analysis of the grounding system measurement data is carried out, where if the grounding resistance value is not based on the PUIL 2011 guidelines, that is, the resistance should be less than 5 Ω, a repair analysis is carried out to obtain a grounding connection according to the standard value [14].

Table 1. Values for 15 Grounding Points at The Semolowaru Feeder Distribution Substation

Point	Rated Earth Resistance (Ω)
BD 510	2.5
BD 311	12
BD 313	1.1
BD 837	8
BD 121	16
BD 317	2
BD 751	3
BD 310	3.5
BD 653	4
BD 650	8
BD 310	16
BD 515	2
BD 518	3
BD 519	3.5
BD 524	4

There are 15 grounding points at the Semolowaru Feeder Distribution Substation, following Table 1 Calculations were made to compare the results of the calculation of the results of the electrode analysis using the manual formula and calculations using ETAP, Figure 1. The results obtained are then whether the land value obtained matches the formula with the ETAP Software.

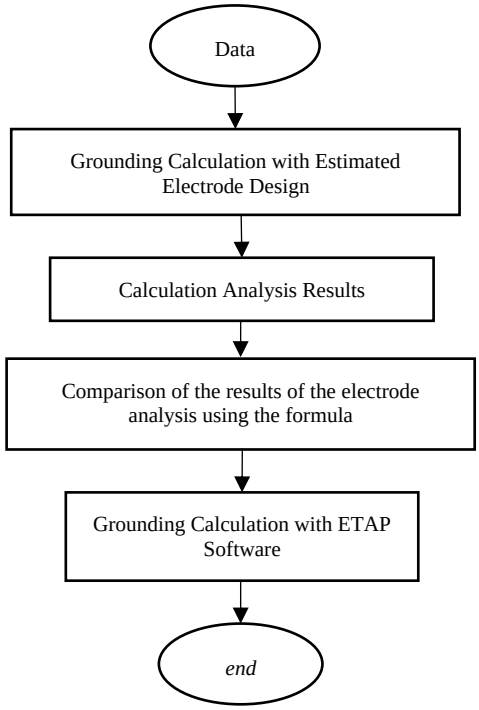


Figure 1. Flowchart Analysis System.

II. RESULTS AND DISCUSSION

The analysis of grounding resistance is carried out in two stages where the first analysis is carried out by mathematical calculations to obtain the value of grounding resistance which should follow the theory and comparison of calculations using the ETAP software. The condition of the soil in the Glagah feeder channel has 1 type of soil, namely swamp land. Calculation of swamp soil conditions.

$$R\ swamp\ land = \frac{\rho\ swamp\ land}{2 \times \pi \times L}\left(\ln\frac{4L}{d} - 1\right) \quad (1)$$

Further analysis is carried out to know the value of the grounding resistance according to the data provided by PT. PLN (Persero) UP3 South Surabaya, namely the Semolowaru feeder. Calculation of the results of the analysis of the value of grounding resistance will be carried out at points in the 20 kV electric voltage distribution line. So, later at certain points where the grounding resistance value is above 5 Ω can be improved by providing a solution so that the grounding resistance value can go down and is less than 5 Ω.

The analysis will be carried out at several points where the ground resistance with the BD series code value does not meet PUIL 2000 requirements which exceed 5 Ω. The following is an analysis of the grounding value at a point

that does not meet the requirements:

A. BD 311 or 2nd Point

Calculation of swamp soil conditions started in equation (2),

$$R\ 2^{nd}\ \text{Point} = \frac{\rho\ 2^{nd}}{2 \times \pi \times L}\left(\ln\frac{4L}{d} - 1\right) \quad (2)$$

After analysing to find out the soil resistivity value in the data Table 1 at point 2 or BD 311, the result is 405.43 Ω/m. So, it can be concluded that with a resistivity value of that size, it must be recalculated to get results below 5 Ω. To get a resistance value below 5 Ω, a re-calculation was carried out while still using a rod electrode with a length of 12m and a diameter of 1.905cm but adding a new electrode with the same size specifications. The second solution is to extend the size of the rod electrode, which was originally 12m to 20m. Here are the two calculations from the two available solutions. based on Table 1 point 4 or BD 837, initially, the grounding resistance value exceeds the requirements, namely 12 Ω, improvements can be made with 2 solutions, namely if you only change the length of the electrode, the grounding resistance value will drop to 5.88 Ω. But if you add a grounding point while still using the same rod electrode size as before, which is still 12m long and 1.905cm in diameter and then connected in parallel, the grounding resistance value will be 0.25 Ω. The second solution is very good because the value of the grounding resistance can be close to zero. The same is the case with the data in Table 1 point 2 or BD 311 and point 4 or BD 837, after an analysis is carried out to find out the soil resistivity value in Table 1 data at point 5 or BD 1212, the result is a value of 540.33 Ω/m. So, it can be concluded that with a resistivity value of that size, it must be recalculated to get results below 5 Ω. To get a resistance value below 5Ω, a re-calculation was carried out while still using a rod electrode with a length of 12m and a diameter of 1.905cm but adding a new electrode with the same size specifications. The second solution is to extend the size of the rod electrode, which was originally 12m to 20m.

B. Analysis of Repairing Grounding Resistance at Qualified Points

This analysis can also be carried out at grounding points that have met the requirements, which is meant by meeting the requirements, namely the value of the grounding resistance is less than 5 Ω. So, by carrying out an analysis other than the point that does not meet the requirements, it can also improve the resistance value so that it can at least reduce it and can be close to zero. There are several similarities in the value of grounding resistance, namely 2.5 Ω, 1.1 Ω, 1.7 Ω, 2 Ω, 3 Ω, 3.4 Ω, 3.5 Ω, 4 Ω, and 2 Ω. Due to the results of the analysis, the improvement in the value of grounding resistance at points that do not meet the requirements, the result is the best if you add electrodes to the grounding system at every point.

Then the new electrode is connected to the old electrode in parallel. So, to improve or meet the requirements can be better in their grounding resistance, alternatively the addition of parallelized rod electrodes will be carried out. The following is an analysis of grounding system points that meet the requirements (3),

$$R_{1,1\,\Omega\,New} = \frac{R_1 \times R_2}{R_1 + R_2} \quad (3)$$

analysis to find out the value of soil resistivity in the data Table 1 at the 6th point or BD 317 and the 10th point or BD 650, we get a value of 1 Ω/m. It can be concluded that with a resistivity value of that size, there is no need to re-calculate because it has obtained results below 5 Ω. indeed the method of adding rod electrodes is connected parallel to the previous electrode. If a resistor is connected in parallel it will reduce the value of the existing resistance so that overvoltage protection such as arresters and ground wires can easily secure overvoltage and can be quickly grounded so as not to damage the components of the electric power distribution and minimize the occurrence of blackouts or blackouts electricity. In the analysis above there is the use of 2 electrodes and there is also a point that uses 3 electrodes so that the resistance value can be reduced. The results of changing the resistance value can be seen in Table 2.

Table 2. Changing in Ground Resistance Value

No.	Value of Grounding Resistance Before Making Changes (Ω)	Earthing Resistance Value After Changes Made (Ω)	Information
1.	1.1	0.55	2 Bar Electrodes
2.	1.7	0.85	
3.	2	1	
4.	2.5	1.25	
5.	3	1.5	3 Bar Electrodes
6.	3.4	1.7	
7.	3.5	1.75	
8.	4	2	

Table 2 shows the results of a significant change in the resistance value where the change in the resistance value is almost close to zero. At a point that has an initial resistance value with a range of 1.1 Ω to 2.5 Ω it can drop from 0.55 Ω to 1.25 Ω. By using 2 electrodes of course. In contrast to the point which has a resistance range of 3 Ω to 4 Ω, after adding 2 rod electrodes and a total of 3 rod electrodes the resistance value can finally drop to 1.5 Ω to 2 Ω. This means that using the method of adding electrodes is very effective in reducing the value of grounding resistance.

C. Comparative Analysis Using ETAP.

Calculation of Step Voltage and Touch Voltage. Land Prisoner The soil fed by Glagah Semolowaru has a swampy soil type with a soil resistance of 30 Ω meters. Coated surface material a thin metal layer of 100 mm (300 Ω meters) is proposed to be installed at that location, so

the Derating factor for the layer (Cs) is contained in formula (4)

$$C_s = 1 - \frac{0{,}09\,(1-\frac{p}{p_s})}{2\times h_s + 0.09} \quad (4)$$

$$C_s = 1 - \frac{0{,}09\,(1-\frac{p}{p_s})}{2\times h_s + 0.09}$$

$$C_s = 0.721$$

The reflection factor is contained in the calculation of formula (5),

$$K = \frac{p-p_s}{p+p_s} \quad (5)$$

$$K = \frac{30-300}{30+300}$$

$$K = \frac{-270}{330}$$

$$K = -0.81$$

The total length of the conductor is contained in the calculation of formula (6)

$$R_g = \frac{p}{4}\sqrt{\frac{\pi}{A}} \quad (6)$$

$$R_g = \frac{30}{4}\sqrt{\frac{0.892}{1250}}$$

$$R_g = 7.5 \times 0.026$$

$$R_g = 0.195$$

The upper limit of earth resistance is contained in the calculation of formula (7),

$$R_g = \frac{p}{4}\sqrt{\frac{\pi}{A}} + \frac{p}{L_T} \quad (7)$$

$$R_g = \frac{30}{4}\sqrt{\frac{0.892}{1250}} + \frac{30}{665}$$

$$R_g = 0.195 + 0.045$$

$$R_g = 0.24$$

Grid depth effect

A square box is 50 m long and 25 m wide, x 6 and y 9 directions. The conductive grid is 200 mm wide and has a depth of 300 mm. There are 40 rods with a length of 3.5 m contained in the calculation of formula (8),

$$R_g = p\left[\frac{1}{L_T} + \frac{1}{\sqrt{20A}}\left(1 + \frac{1}{1+h\sqrt{20/A}}\right)\right] \quad (8)$$

$$R_g = 30\left[\frac{1}{665} + \frac{1}{\sqrt{20.1250}}\left(1 + \frac{1}{1+0.3\sqrt{20/1250}}\right)\right]$$

$$R_g = 30[0.0015 + 0.00017 \times 1.999]$$

$$R_g = 30[0.0015 + 0.00017 \times 1.999]$$

$$R_g = 0.054\ ohm$$

The total length of the conductor embedding is contained in the calculation of formula (9),

$$L_T = L_C + L_R \quad (9)$$

$$L_T = (50 \times 6) + (25 \times 9) + (3.5 \times 40)$$

$$L_T = 300 + 225 + 140$$

$$L_T = 665$$

The number of grid conductors and the length of the grid conductors are contained in the calculation of formula (10),

$$L_C = (N_y + 1)xL_x + (N_x + 1)xL_y \quad (10)$$

$$L_C = ((25 \times 9) + 1) \times 9 + ((50 \times 6) + 1) \times 6$$

$$L_C = 2{,}034 + 1{,}806$$

$$L_C = 3{,}840$$

The length of the conductor is affected by the grid distance contained in the calculation of formula (11)

$$L_C = \left(\left(\frac{6}{1.5} + 1\right)9\right) + \left(\left(\frac{9}{1.5} + 1\right)6\right) \quad (11)$$

$$L_C = 45 + 42$$

$$L_C = 87$$

The total length of the conductor rod is contained in the calculation formula (12),

$$L_R = N_r \times L_r \quad (12)$$

$$L_R = 40 \times 3.5$$

$$L_R = 140$$

The length of the ground rod is affected by the grid contained in the calculation of formula (13),

$$L_R = 2\left(\frac{L_y}{D} + \frac{L_x}{D}\right) \times L_r \quad (13)$$

$$L_R = 2\left(\frac{9}{1.5} + \frac{6}{1.5}\right) \times 3.5$$

$$L_R = 2\left(\frac{9}{1.5} + \frac{6}{1.5}\right) \times 3.5$$

$$L_R = 70$$

Maximum grid current

For example, the maximum single phase to ground fault in the HV winding of a transformer is 3.1 kA and the divide factor is 1 (all fault current goes to ground). The X/R ratio of the disturbance is 15, the maximum duration of the disturbance is 150 ms and the nominal system frequency is 50 Hz. The DC offset time (TA) is contained in the calculation formula (14),

$$T_A = \frac{X}{R} \times \frac{1}{2\pi f} \quad (14)$$

$$T_A = 15 \times \frac{1}{2\pi 50}$$

$$T_A = 0.0477$$

The reduction factor is then contained in the calculation of formula (15)

$$D_f = \sqrt{1 + \frac{T_A}{t_f}\left(1 - e^{\frac{-2f}{T_A}}\right)} \quad (15)$$

$$D_f = \sqrt{1 + \frac{0.0477}{0.15}\left(1 - e^{\frac{-2\times 0.15}{0.0477}}\right)}$$

$$D_f = 1.148$$

Then the maximum grid current is contained in the calculation of formula (16),

$$I_G = I_g D_f \quad (16)$$

$$I_G = 3.1 \times 1.148$$

$$I_G = 3.5585\ KA$$

Criteria for touch stress and step stress

If the average worker's weight is 70 kg, then it is assumed for touch stress and step stress. The maximum allowable touch stress is found in the calculation of formula (17),

$$E_{s_70} = I_B(1000 + 1.5P_s \times C_s) \quad (17)$$

$$E_{s_tol} = 0.58(1000 + 1.5 \times 300 \times 0.721)$$

$$E_{s_tol} = 768{,}181\text{V}$$

The maximum allowable step stress is contained in the calculation of formula (18),

$$E_{l_70} = I_B(1000 + 6P_s \times C_s) \quad (18)$$

$$E_{l_70} = 0.58(1000 + 6 \times 300 \times 0.721)$$

$$E_{l_70} = 1.332{,}72\text{V}$$

So, in the calculation using the formula, it is found that the maximum touch voltage value is 768.181 V and the maximum step voltage is 1332.72 V.

D. Calculations using ETAP Software

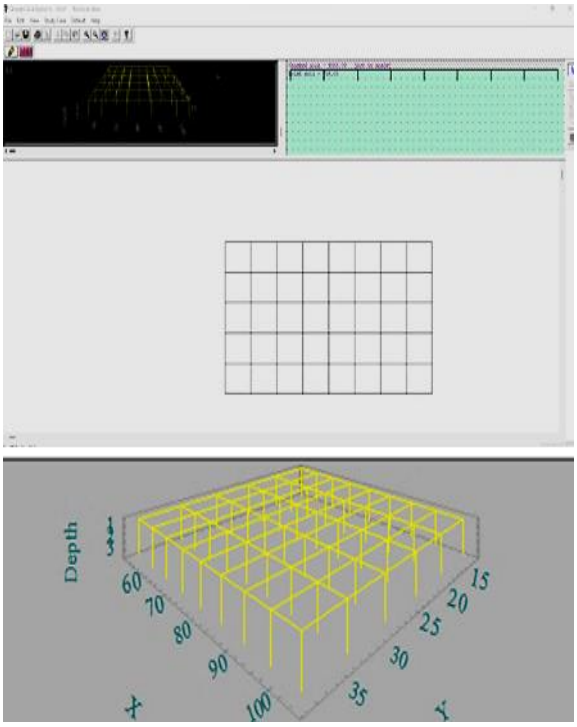


Figure 2. Calculations Using ETAP Software

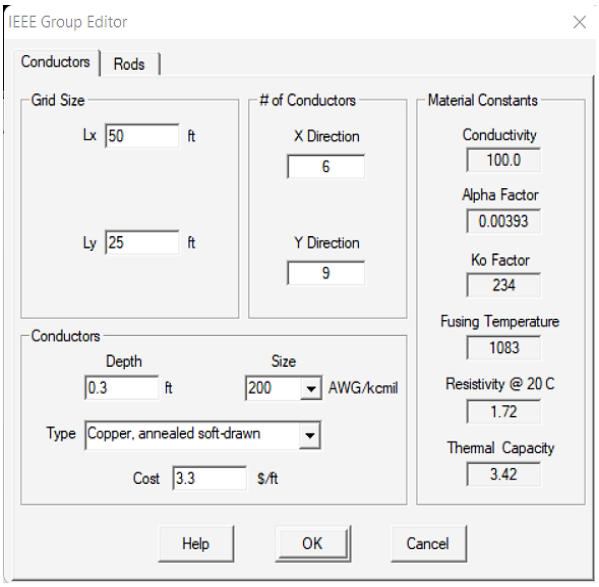


Figure 3. Case Study Conductor

Based on the case study conductor data above, it has a grid size of 50 ft for Lx and 25 ft for Ly. The conductor itself has an x-direction of 6 and a y-direction of 9. The conductor has a depth of 0.3 ft with a size of 200 AWG/kcmil with type copper, annealed soft drawn and a cost value of 3.3 s/ft. The material constants conductivity is 100.0 with an Alpha factor of 0.00393, a ko factor of 234, a fusing temperature of 1083, a resistivity of 20 C of 1.72 and a value of thermal capacity of 3.42.

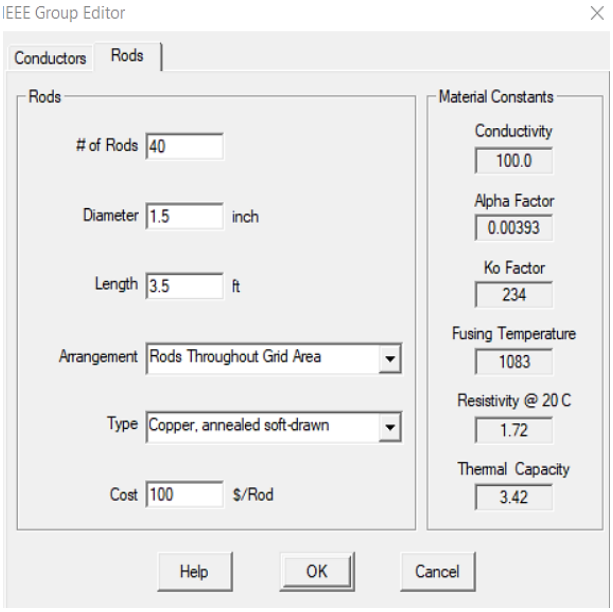


Figure 4. Data of Case Study Rods

Based on the case study rods data above, it has 40 rods, a diameter of 1.5 inches and a length of 3.5 ft. Conductor rods have arrangement rods throughout the grid area with copper type, annealed soft drawn and cost 100 s/rod. The material constants conductivity is 100.0 with an Alpha factor of 0.00393, a ko factor of 234, a fusing temperature of 1083, a resistivity of 20C of 1.72 and a value of thermal capacity of 3.42.

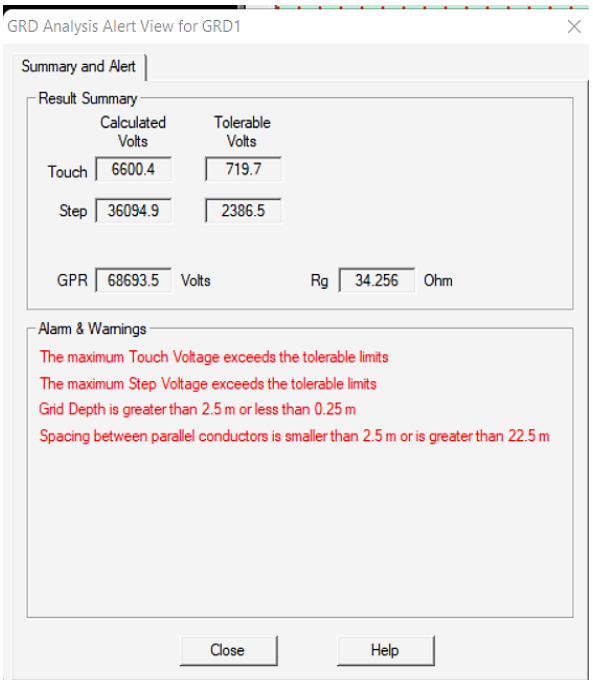


Figure 5. Simulation Results

After doing the simulation calculations, the results obtained are in the touch calculated volts section of 3956.8 while the touch tolerable volts is 587.6. for the result of step calculated volts is 21637.9 while step tolerable volts is 1948.6. the results of the GPR value were 41179.8 volts and Rg was 34.256 Ω. These results get a warning that the maximum touch voltage exceeds the tolerable limit, the maximum step exceeds the tolerable limit, the grid depth is greater than 2.5 m or less than 0.25 m, and the distance between parallel conductors is less than 2.5 m or greater than 22.5 m so it is necessary to correct the data so that there is no red warning.

Comparison using manual calculations and the ETAP application has significantly different results. This difference is due to manual calculations being more focused on grounding depth which refers to PUIL 2011. Meanwhile, calculations using the ETAP application are more focused on Touch tolerable, and Step calculated. In manual calculations it was found that there were different values of grounding resistance before and after changes were made to the resistance values above 5 Ω and below 5 Ω. Whereas in calculations using the ETAP software there are changes to certain values according to the red warning sign on the analysis results.

In addition, there is a comparison of the results for calculating the maximum touch voltage, which is 768.181 volts and for ETAP software calculations for touch tolerable of 587.6 volts, so there is a difference of 180.561 volts. In calculating the maximum step voltage, the result is 1,332.72 volts and in the calculation of the ETAP software, the step tolerable is 1948.6 volts, so there is a difference of 615.88 volts.

Table 2. Changing In Ground Resistance Value

Manual Calculations		**ETAP calculation**			**Calculation of Touch Voltage**
Before Change	**After Change**	**Before Repair**		**after**	
405.43 Ω	0.16 Ω	Conductor	Lx: 50ft	Lx: 50ft, x directions: 3	
269.78 Ω	0.25 Ω		Ly: 25 ft	Ly: 25 ft, y-direction: 6	
540.33 Ω	0.125 Ω		Depth: 0.3 ft	Depth: 1 ft	
1.1 Ω	0.55 Ω	Rods	Rods: 40	Rods: 1146	
1.7 Ω	0.85 Ω		d: 1.5 in	d: 1.5 in	
2 Ω	1 Ω		Length: 3.5 ft	Length: 3.5 ft	
2.5 Ω	1.25 Ω	run	Touch calculated volts: 3956.8	Touch calculated volts: 401.2	
3 Ω	1.5 Ω		Touch tolerable volts: 587.6	Touch tolerable volts: 587.6	Maximum Touch Voltage: 768.181 v
3.4 Ω	1.7 Ω		Step calculated volts: 21637.9	Step calculated volts: 692.7	
3.5 Ω	1.75 Ω		*step tolerable volts: 1948.6*	*step tolerable volts: 1948.6*	Maximum step-up voltage: 1332.72 v
4 Ω	2 Ω		GPR: 41179.8 volts Rg: 34.256 Ω	GPR: 36001.2 volts Rg: 29.948 Ω	

III. CONCLUSION

The condition of the soil in the Semolowaru feeder channel has 1 type of soil, namely swamp land. Of the 15 grounding points analysed, there are 3 points where the grounding value does not meet the requirements of PUIL 2011, namely the grounding value is still above 5 Ω. The three points are the 2nd point with a resistance value of 405.43 Ω, the 4th point with a resistance value of 269.78 Ω, and the 5th point with a resistance value of 540.33 Ω. After analysing using one electrode into 2 pieces and then paralleling it, the result is that the 2nd point has a resistance value of 0.16 Ω, the 4th point has a resistance value of 0.25 Ω, and the 5th point has a resistance value of 0.125 Ω. Analysis using ETAP software was carried out through three analyses, namely the conductor case study, the rods case study, and the run case study. ETAP software, if an error occurs in the calculation, the application will give a warning about what must be

changed so that the calculation is correct. After the conductors and rods are repaired based on the warning given that the maximum touch voltage exceeds the tolerable limit, the maximum stroke exceeds the tolerable limit, the grid depth is greater than 2.5 m or less than 0.25 m, and the distance between parallel conductors is less than 2.5 m or greater than 22.5 m so it is necessary to correct the data so that there is no red warning, the results obtained are correct with the correct steps, Table 3. This paper recommends that a calculation simulation is made using software and a more complete guide so as not to calculate manually. For the points being analysed, a thorough calculation can be carried out so that the overall grounding value of each point in the distribution channel can be known. Completeness of data such as the value of grounding resistance can be more accurate so that the analysis calculations can be more accurate. Grounding analysis can be carried out at the Semolowaru feeder of PT. PLN (Persero) UP3 South Surabaya so that the value of grounding resistance can be analyzed and improvements can also be made so that the value of the grounding system can be better or can be close to zero.

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